

Geography Ch-9 (Must-Do) — Lakes: Formation & Types

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain

Lakes by origin: tectonic, glacial, volcanic, erosional, depositional, ox-bow, lagoon

CA anchor: Chilika Lagoon (Asia's largest brackish lagoon)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Origin & Classification of Lakes (GC Leong / Majid)

GS-I • Physical
Geography →
Lakes (Source:
GC Leong, Majid
Husain)

■ NEWS TRIGGER

Lake geomorphology underpins India's wetland map — from tectonic Wular and lagoonal Chilika to glacial Kashmir tarns — now central to Ramsar conservation.

■ INTRO & ORIGIN

A lake is a body of water in a hollow of the land. GC Leong classifies lakes by the **process that made the hollow** — earth movement, glaciation, volcanic activity, erosion or deposition.

Some lakes fit more than one category, but each is grouped by its dominant formative agent; many are short-lived, being filled by sediment or drained by outlet erosion.

■ KEY DATA

Origin	Lake type	Example
Earth movement	Tectonic / rift-valley lake	Tanganyika, Baikal (rift); Caspian (largest)
Glaciation	Tarn / kettle / drift-dammed	Cirque tarns; Great Lakes (glacial scour)
Volcanic	Crater / caldera lake	Lake Toba (caldera); Lonar (impact/crater)
Erosion	Karst / deflation (playa) lake	Solution lakes; desert playas
Deposition	Ox-bow / lagoon / barrier lake	Cut-off meanders; Chilika (lagoon)
Man-made	Reservoir	Dammed river lakes

■ STATIC THEORY

- **Earth-movement (tectonic) lakes:** warping, sagging or faulting create basins — deep **rift-valley lakes** (Tanganyika, Baikal) and downwarped basins (Caspian, the world's largest lake).
- **Glacial lakes:** ice-scoured hollows (**tarns** in cirques), **kettle lakes** (buried ice blocks melt), and drift-dammed lakes in drumlin lowlands.
- **Volcanic lakes:** water in a **crater/caldera** (Toba) or where a collapsed lava crust leaves a shallow basin (Myvatn, Iceland).
- **Erosional lakes:** solution hollows in karst, and wind-scooped **deflation hollows/playas** in deserts (often salty in inland-drainage basins).
- **Depositional lakes:** **ox-bow lakes** (cut-off meanders), **lagoons** (barrier/spit encloses coastal water — e.g. Chilika), and barrier/beaver-dam lakes.

■ MUST-KNOW FACTS

1. Rift-valley lakes are tectonic & very deep (Tanganyika, Baikal).
2. Caspian Sea = world's largest lake (tectonic/downwarp).
3. Tarn = cirque lake; kettle lake = melted buried ice block.
4. Caldera lake (Toba) vs crater lake; Lonar = impact crater lake.

■ UPSC TRAPS

- ✗ The Caspian Sea is an ocean/sea, not a lake.
- ✓ It is the **WORLD'S LARGEST LAKE** (an inland tectonic basin), despite its name.
- ✗ All deep lakes are glacial.
- ✓ The **DEEPEST** lakes (Baikal, Tanganyika) are **TECTONIC** rift-valley lakes.

5. Ox-bow = cut-off meander; lagoon = barrier-enclosed coastal lake. 6. Salt lakes form in basins of inland (endorheic) drainage.	✗ A lagoon is a tectonic lake. ✓ A lagoon is a DEPOSITIONAL/coastal lake enclosed by a bar or spit (e.g. Chilika).
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■ MAINS ANGLE <i>Classify lakes by mode of origin and apply the scheme to India's tectonic (Wular), glacial (Kashmir), volcanic (Lonar) and lagoonal (Chilika) lakes.</i>	■ STUDY LINK Geography → Physical → Lakes (GC Leong: Lakes; Majid: Lakes of the World)
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■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Chilika Lagoon (Asia's Largest Brackish Lagoon)

GS-I / GS-III · De positional/lagoon lake applied → Wetland conservation

■ NEWS TRIGGER Chilika (Odisha) — Asia's largest brackish-water lagoon and India's first Ramsar site (1981) — is a flagship for lagoon ecology, hosting Irrawaddy dolphins and vast migratory-bird flocks.
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■ INTRO & ORIGIN A lagoon is a coastal depositional lake cut off from the sea by a bar/spit — Chilika is the classic Indian example, brackish because it mixes river and sea water.

■ STATIC THEORY • Chilika is Asia's largest brackish-water lagoon (barrier-enclosed coastal lake) in Odisha. • It was among India's first Ramsar sites (1981) and supports Irrawaddy dolphins and large fisheries. • Its salinity and mouth position are actively managed to sustain fish stocks and bird habitat.

■ MUST-KNOW FACTS 1. Chilika = Asia's largest brackish lagoon (Odisha). 2. India's first Ramsar site (1981, with Keoladeo). 3. Key species: Irrawaddy dolphin; major migratory-bird site.	■ UPSC TRAPS ✗ Chilika is a freshwater tectonic lake. ✓ It is a BRACKISH coastal LAGOON (depositional), mixing river and sea water.
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■ MAINS ANGLE <i>Use lagoon geomorphology to explain Chilika's brackish ecology and its Ramsar conservation stakes.</i>	■ STUDY LINK Geography → Lakes → Lagoons / Wetlands (applied CA)
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